A FUNCTOR BETWEEN DIFFERENTIAL GRADED CATEGORIES WITH AXIOMS (A), (B), (C) INDUCES AN EXACT FUNCTOR

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-dga/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the preservation of admissible short exact sequences (proof line 4716-4718) is the standard additivity argument: a functor of differential graded categories is R-linear, hence additive, hence preserves graded biproducts and commutes with differentials, so termwise-split sequences map to termwise-split sequences; the boundary agreement is computed explicitly in the span; the boundary construction is Lemma lemma-get-triangle (in-source, navigation stub).

Editorial note. The label lemma-functor-between-ABC names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The ABC axioms and the boundary construction. Same unit, Situation situation-ABC and Lemma lemma-get-triangle. The proof uses the boundary morphism delta = pi d(s) attached to an admissible short exact sequence by the construction of Lemma lemma-get-triangle, and the axioms (A), (B), (C) enjoyed by both categories.
- Shift compatibility hypothesis. Hypothesis of the lemma: F(x[1]) = F(x)[1]. The hypothesis identifies the target of F(delta) with F(x)[1], so the image of the boundary morphism lands in the right place.

Source reference. Original reference 09P6: sources/stacks-dga/pinned-original-source.tex, source lines 3708–3708. Prerequisite pointer invoked at proof line(s) 4724; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let R be a ring. Let $F : \mathcal{A} \to \mathcal{B}$ be a functor between differential graded categories over R satisfying axioms (A), (B), and (C) such that $F(x[1]) = F(x)[1]$. Then F induces an exact functor $K(\mathcal{A}) \to K(\mathcal{B})$ of triangulated categories.*

Proof. Namely, if $x \to y \to z$ is an admissible short exact sequence in $\mathrm{Comp}(\mathcal{A})$, then $F(x) \to F(y) \to F(z)$ is an admissible short exact sequence in $\mathrm{Comp}(\mathcal{B})$. Moreover, the "boundary" morphism $\delta = \pi \mathrm{d}(s) : z \to x[1]$ constructed in Lemma 09P6 produces the morphism $F(\delta) : F(z) \to F(x[1]) = F(x)[1]$ which is equal to

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the boundary map $F(\pi)\mathrm{d}(F(s))$ for the admissible short exact sequence $F(x) \to F(y) \to F(z)$. $\square$